PG No. 32 O_{h} $m\bar{3}m$ [cubic]

* Wyckoff site: `6a`, site symmetry: `4m.m`

Table 1: Wyckoff bond: `6a@6a`, bond symmetry: `4m.m`

No.	vector	center	mapping
1	$[X, 0, 0]$	$[x, 0, 0]$	`[1,4,17,20,26,27,42,43]`
2	$[-X, 0, 0]$	$[-x, 0, 0]$	`[2,3,18,19,25,28,41,44]`
3	$[0, X, 0]$	$[0, x, 0]$	`[5,8,13,16,30,31,38,39]`
4	$[0, -X, 0]$	$[0, -x, 0]$	`[6,7,14,15,29,32,37,40]`
5	$[0, 0, X]$	$[0, 0, x]$	`[9,12,22,23,34,35,45,48]`
6	$[0, 0, -X]$	$[0, 0, -x]$	`[10,11,21,24,33,36,46,47]`

Table 2: Wyckoff bond: `12b@6a`, bond symmetry: `2.mm`

No.	vector	center	mapping
1	$[0, X, X]$	$[x, 0, 0]$	`[1,-4,-42,43]`
2	$[0, -X, X]$	$[-x, 0, 0]$	`[2,-3,41,-44]`
3	$[X, 0, X]$	$[0, x, 0]$	`[5,-8,38,-39]`
4	$[X, 0, -X]$	$[0, -x, 0]$	`[6,-7,-37,40]`
5	$[X, X, 0]$	$[0, 0, x]$	`[9,-12,-45,48]`
6	$[-X, X, 0]$	$[0, 0, -x]$	`[10,-11,46,-47]`
7	$[X, 0, -X]$	$[0, x, 0]$	`[13,-16,-30,31]`
8	$[-X, 0, -X]$	$[0, -x, 0]$	`[14,-15,29,-32]`
9	$[0, X, -X]$	$[x, 0, 0]$	`[17,-20,26,-27]`
10	$[0, X, X]$	$[-x, 0, 0]$	`[18,-19,-25,28]`
11	$[X, X, 0]$	$[0, 0, -x]$	`[21,-24,-33,36]`
12	$[X, -X, 0]$	$[0, 0, x]$	`[22,-23,34,-35]`

Table 3: Wyckoff bond: `12c@6a`, bond symmetry: `mm2..`

No.	vector	center	mapping
1	$[0, 0, X]$	$[x, 0, 0]$	`[1,-4,-26,27]`
2	$[0, 0, X]$	$[-x, 0, 0]$	`[2,-3,-25,28]`
3	$[X, 0, 0]$	$[0, x, 0]$	`[5,-8,-30,31]`
4	$[X, 0, 0]$	$[0, -x, 0]$	`[6,-7,-29,32]`
5	$[0, X, 0]$	$[0, 0, x]$	`[9,-12,-34,35]`
6	$[0, X, 0]$	$[0, 0, -x]$	`[10,-11,-33,36]`
7	$[0, 0, -X]$	$[0, x, 0]$	`[13,-16,-38,39]`
8	$[0, 0, -X]$	$[0, -x, 0]$	`[14,-15,-37,40]`
9	$[0, X, 0]$	$[x, 0, 0]$	`[17,-20,-42,43]`
10	$[0, X, 0]$	$[-x, 0, 0]$	`[18,-19,-41,44]`
11	$[X, 0, 0]$	$[0, 0, -x]$	`[21,-24,-46,47]`
12	$[X, 0, 0]$	$[0, 0, x]$	`[22,-23,-45,48]`

Table 4: Wyckoff bond: `24d@6a`, bond symmetry: `..m`

No.	vector	center	mapping
1	$[Y, X, X]$	$[x, 0, 0]$	`[1,43]`
2	$[-Y, -X, X]$	$[-x, 0, 0]$	`[2,41]`
3	$[-Y, X, -X]$	$[-x, 0, 0]$	`[3,44]`
4	$[Y, -X, -X]$	$[x, 0, 0]$	`[4,42]`
5	$[X, Y, X]$	$[0, x, 0]$	`[5,38]`
6	$[X, -Y, -X]$	$[0, -x, 0]$	`[6,40]`
7	$[-X, -Y, X]$	$[0, -x, 0]$	`[7,37]`
8	$[-X, Y, -X]$	$[0, x, 0]$	`[8,39]`
9	$[X, X, Y]$	$[0, 0, x]$	`[9,48]`
10	$[-X, X, -Y]$	$[0, 0, -x]$	`[10,46]`
11	$[X, -X, -Y]$	$[0, 0, -x]$	`[11,47]`
12	$[-X, -X, Y]$	$[0, 0, x]$	`[12,45]`
13	$[X, Y, -X]$	$[0, x, 0]$	`[13,31]`
14	$[-X, -Y, -X]$	$[0, -x, 0]$	`[14,29]`
15	$[X, -Y, X]$	$[0, -x, 0]$	`[15,32]`
16	$[-X, Y, X]$	$[0, x, 0]$	`[16,30]`
17	$[Y, X, -X]$	$[x, 0, 0]$	`[17,26]`
18	$[-Y, X, X]$	$[-x, 0, 0]$	`[18,28]`
19	$[-Y, -X, -X]$	$[-x, 0, 0]$	`[19,25]`
20	$[Y, -X, X]$	$[x, 0, 0]$	`[20,27]`
21	$[X, X, -Y]$	$[0, 0, -x]$	`[21,36]`
22	$[X, -X, Y]$	$[0, 0, x]$	`[22,34]`
23	$[-X, X, Y]$	$[0, 0, x]$	`[23,35]`
24	$[-X, -X, -Y]$	$[0, 0, -x]$	`[24,33]`

Table 5: Wyckoff bond: `24e@6a`, bond symmetry: `m..`

No.	vector	center	mapping
1	$[X, Y, 0]$	$[x, 0, 0]$	`[1,26]`
2	$[-X, -Y, 0]$	$[-x, 0, 0]$	`[2,25]`
3	$[-X, Y, 0]$	$[-x, 0, 0]$	`[3,28]`
4	$[X, -Y, 0]$	$[x, 0, 0]$	`[4,27]`
5	$[0, X, Y]$	$[0, x, 0]$	`[5,30]`
6	$[0, -X, -Y]$	$[0, -x, 0]$	`[6,29]`
7	$[0, -X, Y]$	$[0, -x, 0]$	`[7,32]`
8	$[0, X, -Y]$	$[0, x, 0]$	`[8,31]`
9	$[Y, 0, X]$	$[0, 0, x]$	`[9,34]`
10	$[-Y, 0, -X]$	$[0, 0, -x]$	`[10,33]`
11	$[Y, 0, -X]$	$[0, 0, -x]$	`[11,36]`
12	$[-Y, 0, X]$	$[0, 0, x]$	`[12,35]`
13	$[Y, X, 0]$	$[0, x, 0]$	`[13,38]`
14	$[-Y, -X, 0]$	$[0, -x, 0]$	`[14,37]`
15	$[Y, -X, 0]$	$[0, -x, 0]$	`[15,40]`
16	$[-Y, X, 0]$	$[0, x, 0]$	`[16,39]`
17	$[X, 0, -Y]$	$[x, 0, 0]$	`[17,42]`

continued ...

Table 5

No.	vector	center	mapping
18	$[-X, 0, Y]$	$[-x, 0, 0]$	`[18,41]`
19	$[-X, 0, -Y]$	$[-x, 0, 0]$	`[19,44]`
20	$[X, 0, Y]$	$[x, 0, 0]$	`[20,43]`
21	$[0, Y, -X]$	$[0, 0, -x]$	`[21,46]`
22	$[0, -Y, X]$	$[0, 0, x]$	`[22,45]`
23	$[0, Y, X]$	$[0, 0, x]$	`[23,48]`
24	$[0, -Y, -X]$	$[0, 0, -x]$	`[24,47]`

Table 6: Wyckoff bond: `24f@6a`, bond symmetry: `2..`

No.	vector	center	mapping
1	$[0, X, Y]$	$[x, 0, 0]$	`[1,-4]`
2	$[0, -X, Y]$	$[-x, 0, 0]$	`[2,-3]`
3	$[Y, 0, X]$	$[0, x, 0]$	`[5,-8]`
4	$[Y, 0, -X]$	$[0, -x, 0]$	`[6,-7]`
5	$[X, Y, 0]$	$[0, 0, x]$	`[9,-12]`
6	$[-X, Y, 0]$	$[0, 0, -x]$	`[10,-11]`
7	$[X, 0, -Y]$	$[0, x, 0]$	`[13,-16]`
8	$[-X, 0, -Y]$	$[0, -x, 0]$	`[14,-15]`
9	$[0, Y, -X]$	$[x, 0, 0]$	`[17,-20]`
10	$[0, Y, X]$	$[-x, 0, 0]$	`[18,-19]`
11	$[Y, X, 0]$	$[0, 0, -x]$	`[21,-24]`
12	$[Y, -X, 0]$	$[0, 0, x]$	`[22,-23]`
13	$[0, -X, -Y]$	$[-x, 0, 0]$	`[25,-28]`
14	$[0, X, -Y]$	$[x, 0, 0]$	`[26,-27]`
15	$[-Y, 0, -X]$	$[0, -x, 0]$	`[29,-32]`
16	$[-Y, 0, X]$	$[0, x, 0]$	`[30,-31]`
17	$[-X, -Y, 0]$	$[0, 0, -x]$	`[33,-36]`
18	$[X, -Y, 0]$	$[0, 0, x]$	`[34,-35]`
19	$[-X, 0, Y]$	$[0, -x, 0]$	`[37,-40]`
20	$[X, 0, Y]$	$[0, x, 0]$	`[38,-39]`
21	$[0, -Y, X]$	$[-x, 0, 0]$	`[41,-44]`
22	$[0, -Y, -X]$	$[x, 0, 0]$	`[42,-43]`
23	$[-Y, -X, 0]$	$[0, 0, x]$	`[45,-48]`
24	$[-Y, X, 0]$	$[0, 0, -x]$	`[46,-47]`

Table 7: Wyckoff bond: `48g@6a`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$[x, 0, 0]$	`[1]`
2	$[-X, -Y, Z]$	$[-x, 0, 0]$	`[2]`
3	$[-X, Y, -Z]$	$[-x, 0, 0]$	`[3]`
4	$[X, -Y, -Z]$	$[x, 0, 0]$	`[4]`
5	$[Z, X, Y]$	$[0, x, 0]$	`[5]`

continued ...

Table 7

No.	vector	center	mapping
6	$[Z, -X, -Y]$	$[0, -x, 0]$	[6]
7	$[-Z, -X, Y]$	$[0, -x, 0]$	[7]
8	$[-Z, X, -Y]$	$[0, x, 0]$	[8]
9	$[Y, Z, X]$	$[0, 0, x]$	[9]
10	$[-Y, Z, -X]$	$[0, 0, -x]$	[10]
11	$[Y, -Z, -X]$	$[0, 0, -x]$	[11]
12	$[-Y, -Z, X]$	$[0, 0, x]$	[12]
13	$[Y, X, -Z]$	$[0, x, 0]$	[13]
14	$[-Y, -X, -Z]$	$[0, -x, 0]$	[14]
15	$[Y, -X, Z]$	$[0, -x, 0]$	[15]
16	$[-Y, X, Z]$	$[0, x, 0]$	[16]
17	$[X, Z, -Y]$	$[x, 0, 0]$	[17]
18	$[-X, Z, Y]$	$[-x, 0, 0]$	[18]
19	$[-X, -Z, -Y]$	$[-x, 0, 0]$	[19]
20	$[X, -Z, Y]$	$[x, 0, 0]$	[20]
21	$[Z, Y, -X]$	$[0, 0, -x]$	[21]
22	$[Z, -Y, X]$	$[0, 0, x]$	[22]
23	$[-Z, Y, X]$	$[0, 0, x]$	[23]
24	$[-Z, -Y, -X]$	$[0, 0, -x]$	[24]
25	$[-X, -Y, -Z]$	$[-x, 0, 0]$	[25]
26	$[X, Y, -Z]$	$[x, 0, 0]$	[26]
27	$[X, -Y, Z]$	$[x, 0, 0]$	[27]
28	$[-X, Y, Z]$	$[-x, 0, 0]$	[28]
29	$[-Z, -X, -Y]$	$[0, -x, 0]$	[29]
30	$[-Z, X, Y]$	$[0, x, 0]$	[30]
31	$[Z, X, -Y]$	$[0, x, 0]$	[31]
32	$[Z, -X, Y]$	$[0, -x, 0]$	[32]
33	$[-Y, -Z, -X]$	$[0, 0, -x]$	[33]
34	$[Y, -Z, X]$	$[0, 0, x]$	[34]
35	$[-Y, Z, X]$	$[0, 0, x]$	[35]
36	$[Y, Z, -X]$	$[0, 0, -x]$	[36]
37	$[-Y, -X, Z]$	$[0, -x, 0]$	[37]
38	$[Y, X, Z]$	$[0, x, 0]$	[38]
39	$[-Y, X, -Z]$	$[0, x, 0]$	[39]
40	$[Y, -X, -Z]$	$[0, -x, 0]$	[40]
41	$[-X, -Z, Y]$	$[-x, 0, 0]$	[41]
42	$[X, -Z, -Y]$	$[x, 0, 0]$	[42]
43	$[X, Z, Y]$	$[x, 0, 0]$	[43]
44	$[-X, Z, -Y]$	$[-x, 0, 0]$	[44]
45	$[-Z, -Y, X]$	$[0, 0, x]$	[45]
46	$[-Z, Y, -X]$	$[0, 0, -x]$	[46]
47	$[Z, -Y, -X]$	$[0, 0, -x]$	[47]
48	$[Z, Y, X]$	$[0, 0, x]$	[48]

* Wyckoff site: 8b, site symmetry: .3m

Table 8: Wyckoff bond: `8a@8b`, bond symmetry: `.3m`

No.	vector	center	mapping
1	$[X, X, X]$	$[x, x, x]$	`[1,5,9,38,43,48]`
2	$[-X, -X, X]$	$[-x, -x, x]$	`[2,7,12,37,41,45]`
3	$[-X, X, -X]$	$[-x, x, -x]$	`[3,8,10,39,44,46]`
4	$[X, -X, -X]$	$[x, -x, -x]$	`[4,6,11,40,42,47]`
5	$[X, X, -X]$	$[x, x, -x]$	`[13,17,21,26,31,36]`
6	$[-X, -X, -X]$	$[-x, -x, -x]$	`[14,19,24,25,29,33]`
7	$[X, -X, X]$	$[x, -x, x]$	`[15,20,22,27,32,34]`
8	$[-X, X, X]$	$[-x, x, x]$	`[16,18,23,28,30,35]`

Table 9: Wyckoff bond: `24b@8b`, bond symmetry: `..m`

No.	vector	center	mapping
1	$[X, X, Y]$	$[x, x, x]$	`[1,38]`
2	$[-X, -X, Y]$	$[-x, -x, x]$	`[2,37]`
3	$[-X, X, -Y]$	$[-x, x, -x]$	`[3,39]`
4	$[X, -X, -Y]$	$[x, -x, -x]$	`[4,40]`
5	$[Y, X, X]$	$[x, x, x]$	`[5,48]`
6	$[Y, -X, -X]$	$[x, -x, -x]$	`[6,47]`
7	$[-Y, -X, X]$	$[-x, -x, x]$	`[7,45]`
8	$[-Y, X, -X]$	$[-x, x, -x]$	`[8,46]`
9	$[X, Y, X]$	$[x, x, x]$	`[9,43]`
10	$[-X, Y, -X]$	$[-x, x, -x]$	`[10,44]`
11	$[X, -Y, -X]$	$[x, -x, -x]$	`[11,42]`
12	$[-X, -Y, X]$	$[-x, -x, x]$	`[12,41]`
13	$[X, X, -Y]$	$[x, x, -x]$	`[13,26]`
14	$[-X, -X, -Y]$	$[-x, -x, -x]$	`[14,25]`
15	$[X, -X, Y]$	$[x, -x, x]$	`[15,27]`
16	$[-X, X, Y]$	$[-x, x, x]$	`[16,28]`
17	$[X, Y, -X]$	$[x, x, -x]$	`[17,36]`
18	$[-X, Y, X]$	$[-x, x, x]$	`[18,35]`
19	$[-X, -Y, -X]$	$[-x, -x, -x]$	`[19,33]`
20	$[X, -Y, X]$	$[x, -x, x]$	`[20,34]`
21	$[Y, X, -X]$	$[x, x, -x]$	`[21,31]`
22	$[Y, -X, X]$	$[x, -x, x]$	`[22,32]`
23	$[-Y, X, X]$	$[-x, x, x]$	`[23,30]`
24	$[-Y, -X, -X]$	$[-x, -x, -x]$	`[24,29]`

Table 10: Wyckoff bond: `24c@8b`, bond symmetry: `..m`

No.	vector	center	mapping
1	$[X, -X, 0]$	$[x, x, x]$	`[1,-38]`
2	$[-X, X, 0]$	$[-x, -x, x]$	`[2,-37]`
3	$[-X, -X, 0]$	$[-x, x, -x]$	`[3,-39]`
4	$[X, X, 0]$	$[x, -x, -x]$	`[4,-40]`

continued ...

Table 10

No.	vector	center	mapping
5	$[0, X, -X]$	$[x, x, x]$	`[5,-48]`
6	$[0, -X, X]$	$[x, -x, -x]$	`[6,-47]`
7	$[0, -X, -X]$	$[-x, -x, x]$	`[7,-45]`
8	$[0, X, X]$	$[-x, x, -x]$	`[8,-46]`
9	$[-X, 0, X]$	$[x, x, x]$	`[9,-43]`
10	$[X, 0, -X]$	$[-x, x, -x]$	`[10,-44]`
11	$[-X, 0, -X]$	$[x, -x, -x]$	`[11,-42]`
12	$[X, 0, X]$	$[-x, -x, x]$	`[12,-41]`
13	$[-X, X, 0]$	$[x, x, -x]$	`[13,-26]`
14	$[X, -X, 0]$	$[-x, -x, -x]$	`[14,-25]`
15	$[-X, -X, 0]$	$[x, -x, x]$	`[15,-27]`
16	$[X, X, 0]$	$[-x, x, x]$	`[16,-28]`
17	$[X, 0, X]$	$[x, x, -x]$	`[17,-36]`
18	$[-X, 0, -X]$	$[-x, x, x]$	`[18,-35]`
19	$[-X, 0, X]$	$[-x, -x, -x]$	`[19,-33]`
20	$[X, 0, -X]$	$[x, -x, x]$	`[20,-34]`
21	$[0, -X, -X]$	$[x, x, -x]$	`[21,-31]`
22	$[0, X, X]$	$[x, -x, x]$	`[22,-32]`
23	$[0, -X, X]$	$[-x, x, x]$	`[23,-30]`
24	$[0, X, -X]$	$[-x, -x, -x]$	`[24,-29]`

Table 11: Wyckoff bond: `48d@8b`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$[x, x, x]$	`[1]`
2	$[-X, -Y, Z]$	$[-x, -x, x]$	`[2]`
3	$[-X, Y, -Z]$	$[-x, x, -x]$	`[3]`
4	$[X, -Y, -Z]$	$[x, -x, -x]$	`[4]`
5	$[Z, X, Y]$	$[x, x, x]$	`[5]`
6	$[Z, -X, -Y]$	$[x, -x, -x]$	`[6]`
7	$[-Z, -X, Y]$	$[-x, -x, x]$	`[7]`
8	$[-Z, X, -Y]$	$[-x, x, -x]$	`[8]`
9	$[Y, Z, X]$	$[x, x, x]$	`[9]`
10	$[-Y, Z, -X]$	$[-x, x, -x]$	`[10]`
11	$[Y, -Z, -X]$	$[x, -x, -x]$	`[11]`
12	$[-Y, -Z, X]$	$[-x, -x, x]$	`[12]`
13	$[Y, X, -Z]$	$[x, x, -x]$	`[13]`
14	$[-Y, -X, -Z]$	$[-x, -x, -x]$	`[14]`
15	$[Y, -X, Z]$	$[x, -x, x]$	`[15]`
16	$[-Y, X, Z]$	$[-x, x, x]$	`[16]`
17	$[X, Z, -Y]$	$[x, x, -x]$	`[17]`
18	$[-X, Z, Y]$	$[-x, x, x]$	`[18]`
19	$[-X, -Z, -Y]$	$[-x, -x, -x]$	`[19]`
20	$[X, -Z, Y]$	$[x, -x, x]$	`[20]`
21	$[Z, Y, -X]$	$[x, x, -x]$	`[21]`

continued ...

Table 11

No.	vector	center	mapping
22	$[Z, -Y, X]$	$[x, -x, x]$	[22]
23	$[-Z, Y, X]$	$[-x, x, x]$	[23]
24	$[-Z, -Y, -X]$	$[-x, -x, -x]$	[24]
25	$[-X, -Y, -Z]$	$[-x, -x, -x]$	[25]
26	$[X, Y, -Z]$	$[x, x, -x]$	[26]
27	$[X, -Y, Z]$	$[x, -x, x]$	[27]
28	$[-X, Y, Z]$	$[-x, x, x]$	[28]
29	$[-Z, -X, -Y]$	$[-x, -x, -x]$	[29]
30	$[-Z, X, Y]$	$[-x, x, x]$	[30]
31	$[Z, X, -Y]$	$[x, x, -x]$	[31]
32	$[Z, -X, Y]$	$[x, -x, x]$	[32]
33	$[-Y, -Z, -X]$	$[-x, -x, -x]$	[33]
34	$[Y, -Z, X]$	$[x, -x, x]$	[34]
35	$[-Y, Z, X]$	$[-x, x, x]$	[35]
36	$[Y, Z, -X]$	$[x, x, -x]$	[36]
37	$[-Y, -X, Z]$	$[-x, -x, x]$	[37]
38	$[Y, X, Z]$	$[x, x, x]$	[38]
39	$[-Y, X, -Z]$	$[-x, x, -x]$	[39]
40	$[Y, -X, -Z]$	$[x, -x, -x]$	[40]
41	$[-X, -Z, Y]$	$[-x, -x, x]$	[41]
42	$[X, -Z, -Y]$	$[x, -x, -x]$	[42]
43	$[X, Z, Y]$	$[x, x, x]$	[43]
44	$[-X, Z, -Y]$	$[-x, x, -x]$	[44]
45	$[-Z, -Y, X]$	$[-x, -x, x]$	[45]
46	$[-Z, Y, -X]$	$[-x, x, -x]$	[46]
47	$[Z, -Y, -X]$	$[x, -x, -x]$	[47]
48	$[Z, Y, X]$	$[x, x, x]$	[48]

* Wyckoff site: 12c, site symmetry: m.m2

Table 12: Wyckoff bond: 12a@12c, bond symmetry: m.m2

No.	vector	center	mapping
1	$[0, X, X]$	$[0, y, y]$	[1,18,28,43]
2	$[0, -X, X]$	$[0, -y, y]$	[2,20,27,41]
3	$[0, X, -X]$	$[0, y, -y]$	[3,17,26,44]
4	$[0, -X, -X]$	$[0, -y, -y]$	[4,19,25,42]
5	$[X, 0, X]$	$[y, 0, y]$	[5,15,32,38]
6	$[X, 0, -X]$	$[y, 0, -y]$	[6,13,31,40]
7	$[-X, 0, X]$	$[-y, 0, y]$	[7,16,30,37]
8	$[-X, 0, -X]$	$[-y, 0, -y]$	[8,14,29,39]
9	$[X, X, 0]$	$[y, y, 0]$	[9,21,36,48]
10	$[-X, X, 0]$	$[-y, y, 0]$	[10,23,35,46]
11	$[X, -X, 0]$	$[y, -y, 0]$	[11,22,34,47]
12	$[-X, -X, 0]$	$[-y, -y, 0]$	[12,24,33,45]

Table 13: Wyckoff bond: `12b@12c`, bond symmetry: `m.m2`

No.	vector	center	mapping
1	$[0, X, -X]$	$[0, y, y]$	`[1,-18,28,-43]`
2	$[0, -X, -X]$	$[0, -y, y]$	`[2,-20,27,-41]`
3	$[0, X, X]$	$[0, y, -y]$	`[3,-17,26,-44]`
4	$[0, -X, X]$	$[0, -y, -y]$	`[4,-19,25,-42]`
5	$[-X, 0, X]$	$[y, 0, y]$	`[5,-15,32,-38]`
6	$[-X, 0, -X]$	$[y, 0, -y]$	`[6,-13,31,-40]`
7	$[X, 0, X]$	$[-y, 0, y]$	`[7,-16,30,-37]`
8	$[X, 0, -X]$	$[-y, 0, -y]$	`[8,-14,29,-39]`
9	$[X, -X, 0]$	$[y, y, 0]$	`[9,-21,36,-48]`
10	$[-X, -X, 0]$	$[-y, y, 0]$	`[10,-23,35,-46]`
11	$[X, X, 0]$	$[y, -y, 0]$	`[11,-22,34,-47]`
12	$[-X, X, 0]$	$[-y, -y, 0]$	`[12,-24,33,-45]`

Table 14: Wyckoff bond: `12c@12c`, bond symmetry: `m.m2`

No.	vector	center	mapping
1	$[X, 0, 0]$	$[0, y, y]$	`[1,-18,-28,43]`
2	$[-X, 0, 0]$	$[0, -y, y]$	`[2,-20,-27,41]`
3	$[-X, 0, 0]$	$[0, y, -y]$	`[3,-17,-26,44]`
4	$[X, 0, 0]$	$[0, -y, -y]$	`[4,-19,-25,42]`
5	$[0, X, 0]$	$[y, 0, y]$	`[5,-15,-32,38]`
6	$[0, -X, 0]$	$[y, 0, -y]$	`[6,-13,-31,40]`
7	$[0, -X, 0]$	$[-y, 0, y]$	`[7,-16,-30,37]`
8	$[0, X, 0]$	$[-y, 0, -y]$	`[8,-14,-29,39]`
9	$[0, 0, X]$	$[y, y, 0]$	`[9,-21,-36,48]`
10	$[0, 0, -X]$	$[-y, y, 0]$	`[10,-23,-35,46]`
11	$[0, 0, -X]$	$[y, -y, 0]$	`[11,-22,-34,47]`
12	$[0, 0, X]$	$[-y, -y, 0]$	`[12,-24,-33,45]`

Table 15: Wyckoff bond: `24d@12c`, bond symmetry: `..m`

No.	vector	center	mapping
1	$[Y, X, X]$	$[0, y, y]$	`[1,43]`
2	$[-Y, -X, X]$	$[0, -y, y]$	`[2,41]`
3	$[-Y, X, -X]$	$[0, y, -y]$	`[3,44]`
4	$[Y, -X, -X]$	$[0, -y, -y]$	`[4,42]`
5	$[X, Y, X]$	$[y, 0, y]$	`[5,38]`
6	$[X, -Y, -X]$	$[y, 0, -y]$	`[6,40]`
7	$[-X, -Y, X]$	$[-y, 0, y]$	`[7,37]`
8	$[-X, Y, -X]$	$[-y, 0, -y]$	`[8,39]`
9	$[X, X, Y]$	$[y, y, 0]$	`[9,48]`
10	$[-X, X, -Y]$	$[-y, y, 0]$	`[10,46]`
11	$[X, -X, -Y]$	$[y, -y, 0]$	`[11,47]`
12	$[-X, -X, Y]$	$[-y, -y, 0]$	`[12,45]`

continued ...

Table 15

No.	vector	center	mapping
13	$[X, Y, -X]$	$[y, 0, -y]$	`[13,31]`
14	$[-X, -Y, -X]$	$[-y, 0, -y]$	`[14,29]`
15	$[X, -Y, X]$	$[y, 0, y]$	`[15,32]`
16	$[-X, Y, X]$	$[-y, 0, y]$	`[16,30]`
17	$[Y, X, -X]$	$[0, y, -y]$	`[17,26]`
18	$[-Y, X, X]$	$[0, y, y]$	`[18,28]`
19	$[-Y, -X, -X]$	$[0, -y, -y]$	`[19,25]`
20	$[Y, -X, X]$	$[0, -y, y]$	`[20,27]`
21	$[X, X, -Y]$	$[y, y, 0]$	`[21,36]`
22	$[X, -X, Y]$	$[y, -y, 0]$	`[22,34]`
23	$[-X, X, Y]$	$[-y, y, 0]$	`[23,35]`
24	$[-X, -X, -Y]$	$[-y, -y, 0]$	`[24,33]`

Table 16: Wyckoff bond: `24e@12c`, bond symmetry: `..2`

No.	vector	center	mapping
1	$[Y, X, -X]$	$[0, y, y]$	`[1,-18]`
2	$[-Y, -X, -X]$	$[0, -y, y]$	`[2,-20]`
3	$[-Y, X, X]$	$[0, y, -y]$	`[3,-17]`
4	$[Y, -X, X]$	$[0, -y, -y]$	`[4,-19]`
5	$[-X, Y, X]$	$[y, 0, y]$	`[5,-15]`
6	$[-X, -Y, -X]$	$[y, 0, -y]$	`[6,-13]`
7	$[X, -Y, X]$	$[-y, 0, y]$	`[7,-16]`
8	$[X, Y, -X]$	$[-y, 0, -y]$	`[8,-14]`
9	$[X, -X, Y]$	$[y, y, 0]$	`[9,-21]`
10	$[-X, -X, -Y]$	$[-y, y, 0]$	`[10,-23]`
11	$[X, X, -Y]$	$[y, -y, 0]$	`[11,-22]`
12	$[-X, X, Y]$	$[-y, -y, 0]$	`[12,-24]`
13	$[-Y, -X, X]$	$[0, -y, -y]$	`[25,-42]`
14	$[Y, X, X]$	$[0, y, -y]$	`[26,-44]`
15	$[Y, -X, -X]$	$[0, -y, y]$	`[27,-41]`
16	$[-Y, X, -X]$	$[0, y, y]$	`[28,-43]`
17	$[X, -Y, -X]$	$[-y, 0, -y]$	`[29,-39]`
18	$[X, Y, X]$	$[-y, 0, y]$	`[30,-37]`
19	$[-X, Y, -X]$	$[y, 0, -y]$	`[31,-40]`
20	$[-X, -Y, X]$	$[y, 0, y]$	`[32,-38]`
21	$[-X, X, -Y]$	$[-y, -y, 0]$	`[33,-45]`
22	$[X, X, Y]$	$[y, -y, 0]$	`[34,-47]`
23	$[-X, -X, Y]$	$[-y, y, 0]$	`[35,-46]`
24	$[X, -X, -Y]$	$[y, y, 0]$	`[36,-48]`

Table 17: Wyckoff bond: `24f@12c`, bond symmetry: `m..`

No.	vector	center	mapping
1	$[0, X, Y]$	$[0, y, y]$	`[1,28]`
2	$[0, -X, Y]$	$[0, -y, y]$	`[2,27]`
3	$[0, X, -Y]$	$[0, y, -y]$	`[3,26]`
4	$[0, -X, -Y]$	$[0, -y, -y]$	`[4,25]`
5	$[Y, 0, X]$	$[y, 0, y]$	`[5,32]`
6	$[Y, 0, -X]$	$[y, 0, -y]$	`[6,31]`
7	$[-Y, 0, X]$	$[-y, 0, y]$	`[7,30]`
8	$[-Y, 0, -X]$	$[-y, 0, -y]$	`[8,29]`
9	$[X, Y, 0]$	$[y, y, 0]$	`[9,36]`
10	$[-X, Y, 0]$	$[-y, y, 0]$	`[10,35]`
11	$[X, -Y, 0]$	$[y, -y, 0]$	`[11,34]`
12	$[-X, -Y, 0]$	$[-y, -y, 0]$	`[12,33]`
13	$[X, 0, -Y]$	$[y, 0, -y]$	`[13,40]`
14	$[-X, 0, -Y]$	$[-y, 0, -y]$	`[14,39]`
15	$[X, 0, Y]$	$[y, 0, y]$	`[15,38]`
16	$[-X, 0, Y]$	$[-y, 0, y]$	`[16,37]`
17	$[0, Y, -X]$	$[0, y, -y]$	`[17,44]`
18	$[0, Y, X]$	$[0, y, y]$	`[18,43]`
19	$[0, -Y, -X]$	$[0, -y, -y]$	`[19,42]`
20	$[0, -Y, X]$	$[0, -y, y]$	`[20,41]`
21	$[Y, X, 0]$	$[y, y, 0]$	`[21,48]`
22	$[Y, -X, 0]$	$[y, -y, 0]$	`[22,47]`
23	$[-Y, X, 0]$	$[-y, y, 0]$	`[23,46]`
24	$[-Y, -X, 0]$	$[-y, -y, 0]$	`[24,45]`

Table 18: Wyckoff bond: `48g@12c`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$[0, y, y]$	`[1]`
2	$[-X, -Y, Z]$	$[0, -y, y]$	`[2]`
3	$[-X, Y, -Z]$	$[0, y, -y]$	`[3]`
4	$[X, -Y, -Z]$	$[0, -y, -y]$	`[4]`
5	$[Z, X, Y]$	$[y, 0, y]$	`[5]`
6	$[Z, -X, -Y]$	$[y, 0, -y]$	`[6]`
7	$[-Z, -X, Y]$	$[-y, 0, y]$	`[7]`
8	$[-Z, X, -Y]$	$[-y, 0, -y]$	`[8]`
9	$[Y, Z, X]$	$[y, y, 0]$	`[9]`
10	$[-Y, Z, -X]$	$[-y, y, 0]$	`[10]`
11	$[Y, -Z, -X]$	$[y, -y, 0]$	`[11]`
12	$[-Y, -Z, X]$	$[-y, -y, 0]$	`[12]`
13	$[Y, X, -Z]$	$[y, 0, -y]$	`[13]`
14	$[-Y, -X, -Z]$	$[-y, 0, -y]$	`[14]`
15	$[Y, -X, Z]$	$[y, 0, y]$	`[15]`
16	$[-Y, X, Z]$	$[-y, 0, y]$	`[16]`
17	$[X, Z, -Y]$	$[0, y, -y]$	`[17]`

continued ...

Table 18

No.	vector	center	mapping
18	$[-X, Z, Y]$	$[0, y, y]$	[18]
19	$[-X, -Z, -Y]$	$[0, -y, -y]$	[19]
20	$[X, -Z, Y]$	$[0, -y, y]$	[20]
21	$[Z, Y, -X]$	$[y, y, 0]$	[21]
22	$[Z, -Y, X]$	$[y, -y, 0]$	[22]
23	$[-Z, Y, X]$	$[-y, y, 0]$	[23]
24	$[-Z, -Y, -X]$	$[-y, -y, 0]$	[24]
25	$[-X, -Y, -Z]$	$[0, -y, -y]$	[25]
26	$[X, Y, -Z]$	$[0, y, -y]$	[26]
27	$[X, -Y, Z]$	$[0, -y, y]$	[27]
28	$[-X, Y, Z]$	$[0, y, y]$	[28]
29	$[-Z, -X, -Y]$	$[-y, 0, -y]$	[29]
30	$[-Z, X, Y]$	$[-y, 0, y]$	[30]
31	$[Z, X, -Y]$	$[y, 0, -y]$	[31]
32	$[Z, -X, Y]$	$[y, 0, y]$	[32]
33	$[-Y, -Z, -X]$	$[-y, -y, 0]$	[33]
34	$[Y, -Z, X]$	$[y, -y, 0]$	[34]
35	$[-Y, Z, X]$	$[-y, y, 0]$	[35]
36	$[Y, Z, -X]$	$[y, y, 0]$	[36]
37	$[-Y, -X, Z]$	$[-y, 0, y]$	[37]
38	$[Y, X, Z]$	$[y, 0, y]$	[38]
39	$[-Y, X, -Z]$	$[-y, 0, -y]$	[39]
40	$[Y, -X, -Z]$	$[y, 0, -y]$	[40]
41	$[-X, -Z, Y]$	$[0, -y, y]$	[41]
42	$[X, -Z, -Y]$	$[0, -y, -y]$	[42]
43	$[X, Z, Y]$	$[0, y, y]$	[43]
44	$[-X, Z, -Y]$	$[0, y, -y]$	[44]
45	$[-Z, -Y, X]$	$[-y, -y, 0]$	[45]
46	$[-Z, Y, -X]$	$[-y, y, 0]$	[46]
47	$[Z, -Y, -X]$	$[y, -y, 0]$	[47]
48	$[Z, Y, X]$	$[y, y, 0]$	[48]

* Wyckoff site: 24d, site symmetry: m..

Table 19: Wyckoff bond: 24a@24d, bond symmetry: m..

No.	vector	center	mapping
1	$[0, X, Y]$	$[0, y, z]$	[1,28]
2	$[0, -X, Y]$	$[0, -y, z]$	[2,27]
3	$[0, X, -Y]$	$[0, y, -z]$	[3,26]
4	$[0, -X, -Y]$	$[0, -y, -z]$	[4,25]
5	$[Y, 0, X]$	$[z, 0, y]$	[5,32]
6	$[Y, 0, -X]$	$[z, 0, -y]$	[6,31]
7	$[-Y, 0, X]$	$[-z, 0, y]$	[7,30]
8	$[-Y, 0, -X]$	$[-z, 0, -y]$	[8,29]
9	$[X, Y, 0]$	$[y, z, 0]$	[9,36]
10	$[-X, Y, 0]$	$[-y, z, 0]$	[10,35]

continued ...

Table 19

No.	vector	center	mapping
11	$[X, -Y, 0]$	$[y, -z, 0]$	[11,34]
12	$[-X, -Y, 0]$	$[-y, -z, 0]$	[12,33]
13	$[X, 0, -Y]$	$[y, 0, -z]$	[13,40]
14	$[-X, 0, -Y]$	$[-y, 0, -z]$	[14,39]
15	$[X, 0, Y]$	$[y, 0, z]$	[15,38]
16	$[-X, 0, Y]$	$[-y, 0, z]$	[16,37]
17	$[0, Y, -X]$	$[0, z, -y]$	[17,44]
18	$[0, Y, X]$	$[0, z, y]$	[18,43]
19	$[0, -Y, -X]$	$[0, -z, -y]$	[19,42]
20	$[0, -Y, X]$	$[0, -z, y]$	[20,41]
21	$[Y, X, 0]$	$[z, y, 0]$	[21,48]
22	$[Y, -X, 0]$	$[z, -y, 0]$	[22,47]
23	$[-Y, X, 0]$	$[-z, y, 0]$	[23,46]
24	$[-Y, -X, 0]$	$[-z, -y, 0]$	[24,45]

Table 20: Wyckoff bond: 24b@24d, bond symmetry: m..

No.	vector	center	mapping
1	$[X, 0, 0]$	$[0, y, z]$	[1,-28]
2	$[-X, 0, 0]$	$[0, -y, z]$	[2,-27]
3	$[-X, 0, 0]$	$[0, y, -z]$	[3,-26]
4	$[X, 0, 0]$	$[0, -y, -z]$	[4,-25]
5	$[0, X, 0]$	$[z, 0, y]$	[5,-32]
6	$[0, -X, 0]$	$[z, 0, -y]$	[6,-31]
7	$[0, -X, 0]$	$[-z, 0, y]$	[7,-30]
8	$[0, X, 0]$	$[-z, 0, -y]$	[8,-29]
9	$[0, 0, X]$	$[y, z, 0]$	[9,-36]
10	$[0, 0, -X]$	$[-y, z, 0]$	[10,-35]
11	$[0, 0, -X]$	$[y, -z, 0]$	[11,-34]
12	$[0, 0, X]$	$[-y, -z, 0]$	[12,-33]
13	$[0, X, 0]$	$[y, 0, -z]$	[13,-40]
14	$[0, -X, 0]$	$[-y, 0, -z]$	[14,-39]
15	$[0, -X, 0]$	$[y, 0, z]$	[15,-38]
16	$[0, X, 0]$	$[-y, 0, z]$	[16,-37]
17	$[X, 0, 0]$	$[0, z, -y]$	[17,-44]
18	$[-X, 0, 0]$	$[0, z, y]$	[18,-43]
19	$[-X, 0, 0]$	$[0, -z, -y]$	[19,-42]
20	$[X, 0, 0]$	$[0, -z, y]$	[20,-41]
21	$[0, 0, -X]$	$[z, y, 0]$	[21,-48]
22	$[0, 0, X]$	$[z, -y, 0]$	[22,-47]
23	$[0, 0, X]$	$[-z, y, 0]$	[23,-46]
24	$[0, 0, -X]$	$[-z, -y, 0]$	[24,-45]

Table 21: Wyckoff bond: `48c@24d`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$[0, y, z]$	`[1]`
2	$[-X, -Y, Z]$	$[0, -y, z]$	`[2]`
3	$[-X, Y, -Z]$	$[0, y, -z]$	`[3]`
4	$[X, -Y, -Z]$	$[0, -y, -z]$	`[4]`
5	$[Z, X, Y]$	$[z, 0, y]$	`[5]`
6	$[Z, -X, -Y]$	$[z, 0, -y]$	`[6]`
7	$[-Z, -X, Y]$	$[-z, 0, y]$	`[7]`
8	$[-Z, X, -Y]$	$[-z, 0, -y]$	`[8]`
9	$[Y, Z, X]$	$[y, z, 0]$	`[9]`
10	$[-Y, Z, -X]$	$[-y, z, 0]$	`[10]`
11	$[Y, -Z, -X]$	$[y, -z, 0]$	`[11]`
12	$[-Y, -Z, X]$	$[-y, -z, 0]$	`[12]`
13	$[Y, X, -Z]$	$[y, 0, -z]$	`[13]`
14	$[-Y, -X, -Z]$	$[-y, 0, -z]$	`[14]`
15	$[Y, -X, Z]$	$[y, 0, z]$	`[15]`
16	$[-Y, X, Z]$	$[-y, 0, z]$	`[16]`
17	$[X, Z, -Y]$	$[0, z, -y]$	`[17]`
18	$[-X, Z, Y]$	$[0, z, y]$	`[18]`
19	$[-X, -Z, -Y]$	$[0, -z, -y]$	`[19]`
20	$[X, -Z, Y]$	$[0, -z, y]$	`[20]`
21	$[Z, Y, -X]$	$[z, y, 0]$	`[21]`
22	$[Z, -Y, X]$	$[z, -y, 0]$	`[22]`
23	$[-Z, Y, X]$	$[-z, y, 0]$	`[23]`
24	$[-Z, -Y, -X]$	$[-z, -y, 0]$	`[24]`
25	$[-X, -Y, -Z]$	$[0, -y, -z]$	`[25]`
26	$[X, Y, -Z]$	$[0, y, -z]$	`[26]`
27	$[X, -Y, Z]$	$[0, -y, z]$	`[27]`
28	$[-X, Y, Z]$	$[0, y, z]$	`[28]`
29	$[-Z, -X, -Y]$	$[-z, 0, -y]$	`[29]`
30	$[-Z, X, Y]$	$[-z, 0, y]$	`[30]`
31	$[Z, X, -Y]$	$[z, 0, -y]$	`[31]`
32	$[Z, -X, Y]$	$[z, 0, y]$	`[32]`
33	$[-Y, -Z, -X]$	$[-y, -z, 0]$	`[33]`
34	$[Y, -Z, X]$	$[y, -z, 0]$	`[34]`
35	$[-Y, Z, X]$	$[-y, z, 0]$	`[35]`
36	$[Y, Z, -X]$	$[y, z, 0]$	`[36]`
37	$[-Y, -X, Z]$	$[-y, 0, z]$	`[37]`
38	$[Y, X, Z]$	$[y, 0, z]$	`[38]`
39	$[-Y, X, -Z]$	$[-y, 0, -z]$	`[39]`
40	$[Y, -X, -Z]$	$[y, 0, -z]$	`[40]`
41	$[-X, -Z, Y]$	$[0, -z, y]$	`[41]`
42	$[X, -Z, -Y]$	$[0, -z, -y]$	`[42]`
43	$[X, Z, Y]$	$[0, z, y]$	`[43]`
44	$[-X, Z, -Y]$	$[0, z, -y]$	`[44]`
45	$[-Z, -Y, X]$	$[-z, -y, 0]$	`[45]`
46	$[-Z, Y, -X]$	$[-z, y, 0]$	`[46]`

continued ...

Table 21

No.	vector	center	mapping
47	$[Z, -Y, -X]$	$[z, -y, 0]$	[47]
48	$[Z, Y, X]$	$[z, y, 0]$	[48]

* Wyckoff site: 24e, site symmetry: ..m

Table 22: Wyckoff bond: 24a@24e, bond symmetry: ..m

No.	vector	center	mapping
1	$[X, X, Y]$	$[x, x, z]$	[1,38]
2	$[-X, -X, Y]$	$[-x, -x, z]$	[2,37]
3	$[-X, X, -Y]$	$[-x, x, -z]$	[3,39]
4	$[X, -X, -Y]$	$[x, -x, -z]$	[4,40]
5	$[Y, X, X]$	$[z, x, x]$	[5,48]
6	$[Y, -X, -X]$	$[z, -x, -x]$	[6,47]
7	$[-Y, -X, X]$	$[-z, -x, x]$	[7,45]
8	$[-Y, X, -X]$	$[-z, x, -x]$	[8,46]
9	$[X, Y, X]$	$[x, z, x]$	[9,43]
10	$[-X, Y, -X]$	$[-x, z, -x]$	[10,44]
11	$[X, -Y, -X]$	$[x, -z, -x]$	[11,42]
12	$[-X, -Y, X]$	$[-x, -z, x]$	[12,41]
13	$[X, X, -Y]$	$[x, x, -z]$	[13,26]
14	$[-X, -X, -Y]$	$[-x, -x, -z]$	[14,25]
15	$[X, -X, Y]$	$[x, -x, z]$	[15,27]
16	$[-X, X, Y]$	$[-x, x, z]$	[16,28]
17	$[X, Y, -X]$	$[x, z, -x]$	[17,36]
18	$[-X, Y, X]$	$[-x, z, x]$	[18,35]
19	$[-X, -Y, -X]$	$[-x, -z, -x]$	[19,33]
20	$[X, -Y, X]$	$[x, -z, x]$	[20,34]
21	$[Y, X, -X]$	$[z, x, -x]$	[21,31]
22	$[Y, -X, X]$	$[z, -x, x]$	[22,32]
23	$[-Y, X, X]$	$[-z, x, x]$	[23,30]
24	$[-Y, -X, -X]$	$[-z, -x, -x]$	[24,29]

Table 23: Wyckoff bond: 24b@24e, bond symmetry: ..m

No.	vector	center	mapping
1	$[X, -X, 0]$	$[x, x, z]$	[1,-38]
2	$[-X, X, 0]$	$[-x, -x, z]$	[2,-37]
3	$[-X, -X, 0]$	$[-x, x, -z]$	[3,-39]
4	$[X, X, 0]$	$[x, -x, -z]$	[4,-40]
5	$[0, X, -X]$	$[z, x, x]$	[5,-48]
6	$[0, -X, X]$	$[z, -x, -x]$	[6,-47]
7	$[0, -X, -X]$	$[-z, -x, x]$	[7,-45]
8	$[0, X, X]$	$[-z, x, -x]$	[8,-46]
9	$[-X, 0, X]$	$[x, z, x]$	[9,-43]
10	$[X, 0, -X]$	$[-x, z, -x]$	[10,-44]

continued ...

Table 23

No.	vector	center	mapping
11	$[-X, 0, -X]$	$[x, -z, -x]$	`[11,-42]`
12	$[X, 0, X]$	$[-x, -z, x]$	`[12,-41]`
13	$[-X, X, 0]$	$[x, x, -z]$	`[13,-26]`
14	$[X, -X, 0]$	$[-x, -x, -z]$	`[14,-25]`
15	$[-X, -X, 0]$	$[x, -x, z]$	`[15,-27]`
16	$[X, X, 0]$	$[-x, x, z]$	`[16,-28]`
17	$[X, 0, X]$	$[x, z, -x]$	`[17,-36]`
18	$[-X, 0, -X]$	$[-x, z, x]$	`[18,-35]`
19	$[-X, 0, X]$	$[-x, -z, -x]$	`[19,-33]`
20	$[X, 0, -X]$	$[x, -z, x]$	`[20,-34]`
21	$[0, -X, -X]$	$[z, x, -x]$	`[21,-31]`
22	$[0, X, X]$	$[z, -x, x]$	`[22,-32]`
23	$[0, -X, X]$	$[-z, x, x]$	`[23,-30]`
24	$[0, X, -X]$	$[-z, -x, -x]$	`[24,-29]`

Table 24: Wyckoff bond: `48c@24e`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$[x, x, z]$	`[1]`
2	$[-X, -Y, Z]$	$[-x, -x, z]$	`[2]`
3	$[-X, Y, -Z]$	$[-x, x, -z]$	`[3]`
4	$[X, -Y, -Z]$	$[x, -x, -z]$	`[4]`
5	$[Z, X, Y]$	$[z, x, x]$	`[5]`
6	$[Z, -X, -Y]$	$[z, -x, -x]$	`[6]`
7	$[-Z, -X, Y]$	$[-z, -x, x]$	`[7]`
8	$[-Z, X, -Y]$	$[-z, x, -x]$	`[8]`
9	$[Y, Z, X]$	$[x, z, x]$	`[9]`
10	$[-Y, Z, -X]$	$[-x, z, -x]$	`[10]`
11	$[Y, -Z, -X]$	$[x, -z, -x]$	`[11]`
12	$[-Y, -Z, X]$	$[-x, -z, x]$	`[12]`
13	$[Y, X, -Z]$	$[x, x, -z]$	`[13]`
14	$[-Y, -X, -Z]$	$[-x, -x, -z]$	`[14]`
15	$[Y, -X, Z]$	$[x, -x, z]$	`[15]`
16	$[-Y, X, Z]$	$[-x, x, z]$	`[16]`
17	$[X, Z, -Y]$	$[x, z, -x]$	`[17]`
18	$[-X, Z, Y]$	$[-x, z, x]$	`[18]`
19	$[-X, -Z, -Y]$	$[-x, -z, -x]$	`[19]`
20	$[X, -Z, Y]$	$[x, -z, x]$	`[20]`
21	$[Z, Y, -X]$	$[z, x, -x]$	`[21]`
22	$[Z, -Y, X]$	$[z, -x, x]$	`[22]`
23	$[-Z, Y, X]$	$[-z, x, x]$	`[23]`
24	$[-Z, -Y, -X]$	$[-z, -x, -x]$	`[24]`
25	$[-X, -Y, -Z]$	$[-x, -x, -z]$	`[25]`
26	$[X, Y, -Z]$	$[x, x, -z]$	`[26]`
27	$[X, -Y, Z]$	$[x, -x, z]$	`[27]`

continued ...

Table 24

No.	vector	center	mapping
28	$[-X, Y, Z]$	$[-x, x, z]$	[28]
29	$[-Z, -X, -Y]$	$[-z, -x, -x]$	[29]
30	$[-Z, X, Y]$	$[-z, x, x]$	[30]
31	$[Z, X, -Y]$	$[z, x, -x]$	[31]
32	$[Z, -X, Y]$	$[z, -x, x]$	[32]
33	$[-Y, -Z, -X]$	$[-x, -z, -x]$	[33]
34	$[Y, -Z, X]$	$[x, -z, x]$	[34]
35	$[-Y, Z, X]$	$[-x, z, x]$	[35]
36	$[Y, Z, -X]$	$[x, z, -x]$	[36]
37	$[-Y, -X, Z]$	$[-x, -x, z]$	[37]
38	$[Y, X, Z]$	$[x, x, z]$	[38]
39	$[-Y, X, -Z]$	$[-x, x, -z]$	[39]
40	$[Y, -X, -Z]$	$[x, -x, -z]$	[40]
41	$[-X, -Z, Y]$	$[-x, -z, x]$	[41]
42	$[X, -Z, -Y]$	$[x, -z, -x]$	[42]
43	$[X, Z, Y]$	$[x, z, x]$	[43]
44	$[-X, Z, -Y]$	$[-x, z, -x]$	[44]
45	$[-Z, -Y, X]$	$[-z, -x, x]$	[45]
46	$[-Z, Y, -X]$	$[-z, x, -x]$	[46]
47	$[Z, -Y, -X]$	$[z, -x, -x]$	[47]
48	$[Z, Y, X]$	$[z, x, x]$	[48]

* Wyckoff site: 48f, site symmetry: 1

Table 25: Wyckoff bond: 48a@48f, bond symmetry: 1

No.	vector	center	mapping
1	$[X, Y, Z]$	$[x, y, z]$	[1]
2	$[-X, -Y, Z]$	$[-x, -y, z]$	[2]
3	$[-X, Y, -Z]$	$[-x, y, -z]$	[3]
4	$[X, -Y, -Z]$	$[x, -y, -z]$	[4]
5	$[Z, X, Y]$	$[z, x, y]$	[5]
6	$[Z, -X, -Y]$	$[z, -x, -y]$	[6]
7	$[-Z, -X, Y]$	$[-z, -x, y]$	[7]
8	$[-Z, X, -Y]$	$[-z, x, -y]$	[8]
9	$[Y, Z, X]$	$[y, z, x]$	[9]
10	$[-Y, Z, -X]$	$[-y, z, -x]$	[10]
11	$[Y, -Z, -X]$	$[y, -z, -x]$	[11]
12	$[-Y, -Z, X]$	$[-y, -z, x]$	[12]
13	$[Y, X, -Z]$	$[y, x, -z]$	[13]
14	$[-Y, -X, -Z]$	$[-y, -x, -z]$	[14]
15	$[Y, -X, Z]$	$[y, -x, z]$	[15]
16	$[-Y, X, Z]$	$[-y, x, z]$	[16]
17	$[X, Z, -Y]$	$[x, z, -y]$	[17]
18	$[-X, Z, Y]$	$[-x, z, y]$	[18]
19	$[-X, -Z, -Y]$	$[-x, -z, -y]$	[19]
20	$[X, -Z, Y]$	$[x, -z, y]$	[20]

continued ...

Table 25

No.	vector	center	mapping
21	$[Z, Y, -X]$	$[z, y, -x]$	[21]
22	$[Z, -Y, X]$	$[z, -y, x]$	[22]
23	$[-Z, Y, X]$	$[-z, y, x]$	[23]
24	$[-Z, -Y, -X]$	$[-z, -y, -x]$	[24]
25	$[-X, -Y, -Z]$	$[-x, -y, -z]$	[25]
26	$[X, Y, -Z]$	$[x, y, -z]$	[26]
27	$[X, -Y, Z]$	$[x, -y, z]$	[27]
28	$[-X, Y, Z]$	$[-x, y, z]$	[28]
29	$[-Z, -X, -Y]$	$[-z, -x, -y]$	[29]
30	$[-Z, X, Y]$	$[-z, x, y]$	[30]
31	$[Z, X, -Y]$	$[z, x, -y]$	[31]
32	$[Z, -X, Y]$	$[z, -x, y]$	[32]
33	$[-Y, -Z, -X]$	$[-y, -z, -x]$	[33]
34	$[Y, -Z, X]$	$[y, -z, x]$	[34]
35	$[-Y, Z, X]$	$[-y, z, x]$	[35]
36	$[Y, Z, -X]$	$[y, z, -x]$	[36]
37	$[-Y, -X, Z]$	$[-y, -x, z]$	[37]
38	$[Y, X, Z]$	$[y, x, z]$	[38]
39	$[-Y, X, -Z]$	$[-y, x, -z]$	[39]
40	$[Y, -X, -Z]$	$[y, -x, -z]$	[40]
41	$[-X, -Z, Y]$	$[-x, -z, y]$	[41]
42	$[X, -Z, -Y]$	$[x, -z, -y]$	[42]
43	$[X, Z, Y]$	$[x, z, y]$	[43]
44	$[-X, Z, -Y]$	$[-x, z, -y]$	[44]
45	$[-Z, -Y, X]$	$[-z, -y, x]$	[45]
46	$[-Z, Y, -X]$	$[-z, y, -x]$	[46]
47	$[Z, -Y, -X]$	$[z, -y, -x]$	[47]
48	$[Z, Y, X]$	$[z, y, x]$	[48]